IBM Quantum Safe Remediator

Version 1.1.2.1

Transform your enterprise with quantum-safe cryptography and build crypto-agility.

Highlights
Adaptive Proxy – Enable crypto-agility with limited changes

Adaptive Proxy Manager – Unified adaptive proxy monitoring

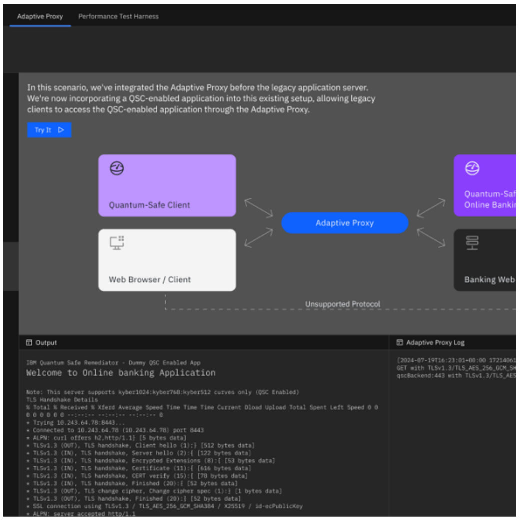
Forward Proxy – Securing outbound traffic

Performance Harness – Free insights before actions

Quantum computing is no longer a distant concept—it's a looming cybersecurity threat. Cybercriminals are likely to be engaging in "harvest now, decrypt later" attacks, collecting encrypted data today in anticipation of breaking it with quantum capabilities when cryptographically relevant quantum computing becomes available. The transition to quantum-safe cryptography (QSC) has become a business and leadership imperative.

For enterprises with complex IT environment, heterogeneous environments, and mission critical legacy applications, transitioning to quantum-safe cryptography presents a unique set of challenges. These environments often cannot be easily updated, and introducing changes to application code or network configurations carries risk and cost. Organizations need a solution that protects their infrastructure today—without disruptions.

IBM Quantum Safe™ Remediator helps organizations strengthen cyber resiliency and move toward quantum-safe cryptography, even when existing application code cannot be changed—an essential first step for modernizing legacy or vendor-managed systems. It enables hybrid cryptographic environments that support both classical and post-quantum algorithms, defends against "harvest now, decrypt later" attacks, and allows teams to test remediation patterns while evaluating algorithm performance in real-world conditions. With centralized cryptography management and seamless infrastructure integration, IBM Quantum Safe Remediator helps modernize security—ensuring enterprises can upgrade to using quantum-safe cryptography with agility, efficiency, and confidence.



Adaptive Proxy – Enable crypto-agility with limited changes

Adaptive Proxy helps organizations enable crypto-agility with limited need to rewrite application code. It supports interoperability across legacy, hybrid, and quantum-safe environments while providing centralized control over TLS protocols and policies. Adaptive Proxy also supports domain-based configuration for VM-based deployments, making it easier to manage adaptive routing, certificate generation, domain aliases, and multi-tier TLS configurations across protected applications. This helps teams accelerate deployment and simplify operational management while defending against “harvest now, decrypt later” threats. It also supports flexible deployment across standalone virtual machines, IBM Cloud Kubernetes Service and Redhat OpenShift on IBM Cloud. For resource-constrained environments such as IoT, Adaptive Proxy can help protect information in transit without requiring devices to implement PQC directly.

Adaptive Proxy Manager – Unified adaptive proxy monitoring

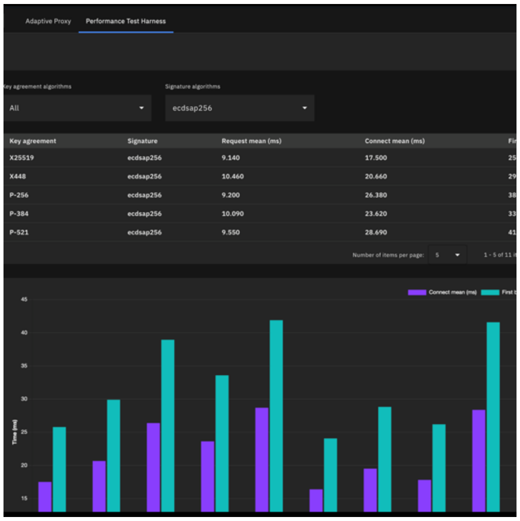
IBM Quantum Safe Remediator Adaptive Proxy Manager provides a centralized interface for administrators to monitor Adaptive Proxy instances, view configuration and health status, and perform selective configuration updates. It also supports the new domain-based configuration model for standalone Adaptive Proxy deployments, helping teams manage proxy applications and configuration changes through a single interface.

Forward Proxy – Securing outbound traffic

The Forward Proxy acts as a critical intermediary, securing outgoing traffic between clients and the internet. It intercepts TLS requests, ensuring that all data leaving the network is encrypted. This is essential when parts of the supply chain aren’t yet quantum-safe, providing an added layer of protection against quantum threats. For example, financial institutions can secure sensitive transactions to organizations with non-compliant supply chain partners and still ensure their communications remain safe by encrypting outgoing traffic with the NIST PQC algorithms before it reaches third-party systems. Forward Proxy ensures that your outgoing traffic is always quantum-safe, no matter where it’s going. The client app adds flexibility, allowing users to turn the proxy on or off as needed, depending on their security environment.

PQC readiness across the solution

IBM Quantum Safe Remediator enables post-quantum cryptography readiness across all core components, helping organizations strengthen cryptographic agility, support security compliance, and operate quantum-safe protections more consistently across inbound traffic, outbound traffic, management workflows, and performance testing.



Performance Harness – Free insights before actions

Performance Harness enables the enterprise to measure HTTP and HTTP/2 traffic and allow users to understand how the algorithms are impacting behaviors. The analysis can help the enterprise take actions that optimize the performance of cryptographic algorithms in both client infrastructure and cloud environments. By running tests using OpenSSL and the benchmarking tool h2load, the Performance Harness allows a security infrastructure architect to gather specific environment performance metrics and compare them to standardized benchmarks, providing valuable insights that guide your quantum-safe transformation efforts.
With a dashboard view, users can benchmark various algorithm combinations and select from a range of test scenarios to run performance tests in the environment. The results, stored in a file system, can be compared to determine how different cryptographic algorithms impact performance. This data-driven approach improves the enterprise transition to using quantum-safe cryptography.

IBM Quantum Safe Remediator pre-requisites:

	Adaptive Proxy / Adaptive Proxy Manager / Performance Harness	**Forward Proxy**
System Requirements	**Virtual Server Instance / Virtual Machine** CPU: 8-core CPU or similar, CPU 2 for Adaptive Proxy Manager RAM: 64GB, 4GB for Adaptive Proxy Manager Free disk space: 10GB, 100GB for Adaptive Proxy Manager **Operating System** Red Hat Enterprise Linux v7.1 or later Ubuntu v20.04 or later	**Operating System** Windows 11 and above Linux - Ubuntu
Software Requirements	Docker Engine v25.0.3 or Later Web browser - Chrome, Safari, Firefox or Edge Internet connection will be needed for Adaptive Proxy Manager VPN connection may be needed for Adaptive Proxy Manager, as per the requirement of the target environment.	PowerShell for Windows /Bash shell for Linux Podman client v4.9.3 or above (for Windows/Linux) or Docker Client v25.0.3 or above
Red Hat Open Shift (RHOS)/IBM Cloud Kubernetes Service requirements	**Cluster deployment for Adaptive Proxy only** Kubernetes v1.30 or above OpenShift client version 4.15.0 or later CPU 500 millicore for each running pod RAM 128 MB per running pod Helm v3.15.2 or later	

IBM Quantum Safe Remediator helps enterprises transition to quantum-safe cryptography with minimal disruption by enabling hybrid environments, securing communications, and defending against future decryption threats. It delivers crypto-agility while preserving operational stability, making modernization both secure and seamless.

For more information

To learn more about our quantum-safe data security solutions, contact your IBM representative or IBM Business Partner, or visit ibm.com/products/guardium-cryptography-manager.

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